

6.6 The Frisch–Waugh–Lovell theorem

A coefficient in a multiple regression is often described as the effect of its regressor “holding the other regressors fixed”. In observational data nobody held anything fixed, so the phrase needs a precise meaning. The Frisch–Waugh–Lovell (FWL) theorem supplies one. A coefficient in a multiple regression equals the slope of a simple regression in which the other regressors have been *projected out* of both the response and the regressor of interest.

6.6.1 Statement and proof

Partition the model matrix and coefficients as

$$\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2], \quad \boldsymbol{\beta} = \begin{pmatrix} \boldsymbol{\beta}_1 \\ \boldsymbol{\beta}_2 \end{pmatrix},$$

with $\mathbf{X}_1$ of size $n \times p_1$ and $\mathbf{X}_2$ of size $n \times p_2$, and assume $\mathbf{X}$ has full column rank. Let $\mathbf{M}_2$ project onto $\mathcal{C}(\mathbf{X}_2)$, and write

$$\tilde{\mathbf{X}}_1 = (\mathbf{I} - \mathbf{M}_2)\mathbf{X}_1, \quad \tilde{\mathbf{y}} = (\mathbf{I} - \mathbf{M}_2)\mathbf{y}$$

for the parts of $\mathbf{X}_1$ and $\mathbf{y}$ that $\mathbf{X}_2$ cannot explain linearly. Each column of $\tilde{\mathbf{X}}_1$ is the residual vector from regressing the corresponding column of $\mathbf{X}_1$ on $\mathbf{X}_2$.

Lemma 6.6.1 — Orthogonalizing a partitioned model space

Under the assumptions above:

- (a) $\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{X}_2) \oplus \mathcal{C}(\tilde{\mathbf{X}}_1)$, and the two summands are orthogonal;
- (b) $\tilde{\mathbf{X}}_1$ has full column rank;
- (c) $\mathbf{M} = \mathbf{M}_2 + \tilde{\mathbf{M}}_1$, where $\tilde{\mathbf{M}}_1 = \tilde{\mathbf{X}}_1(\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top$.

Proof.

- (a) The columns of $\tilde{\mathbf{X}}_1 = \mathbf{X}_1 - \mathbf{M}_2\mathbf{X}_1$ lie in $\mathcal{C}(\mathbf{X})$ and are orthogonal to $\mathcal{C}(\mathbf{X}_2)$. Conversely, for any $\mathbf{b} = (\mathbf{b}_1^\top, \mathbf{b}_2^\top)^\top$,

$$\mathbf{X}\mathbf{b} = \mathbf{X}_1\mathbf{b}_1 + \mathbf{X}_2\mathbf{b}_2 = \tilde{\mathbf{X}}_1\mathbf{b}_1 + (\mathbf{M}_2\mathbf{X}_1\mathbf{b}_1 + \mathbf{X}_2\mathbf{b}_2),$$

with the second term in $\mathcal{C}(\mathbf{X}_2)$. So $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\tilde{\mathbf{X}}_1) + \mathcal{C}(\mathbf{X}_2)$.

- (b) If $\tilde{\mathbf{X}}_1\mathbf{a} = \mathbf{0}$, then $\mathbf{X}_1\mathbf{a} = \mathbf{M}_2\mathbf{X}_1\mathbf{a} = \mathbf{X}_2\mathbf{c}$ for some $\mathbf{c}$, so $\mathbf{X}(\mathbf{a}^\top, -\mathbf{c}^\top)^\top = \mathbf{0}$. Full column rank of $\mathbf{X}$ forces $\mathbf{a} = \mathbf{0}$.
- (c) Combine (a) with Theorem 6.3.8 and Theorem 6.3.7.

■

Theorem 6.6.2 — Frisch–Waugh–Lovell

Let $\hat{\boldsymbol{\beta}} = (\hat{\boldsymbol{\beta}}_1^\top, \hat{\boldsymbol{\beta}}_2^\top)^\top$ be the least squares estimate in the full model. Then:

- (a) $\hat{\beta}_1 = (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{y}} = (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top \mathbf{y}$. So $\hat{\beta}_1$ is the coefficient from regressing $\tilde{\mathbf{y}}$ (or $\mathbf{y}$ itself) on $\tilde{\mathbf{X}}_1$;
- (b) the residual vector from regressing $\tilde{\mathbf{y}}$ on $\tilde{\mathbf{X}}_1$ equals the residual vector $(\mathbf{I} - \mathbf{M})\mathbf{y}$ of the full model;
- (c) $\hat{\beta}_2 = (\mathbf{X}_2^\top \mathbf{X}_2)^{-1} \mathbf{X}_2^\top (\mathbf{y} - \mathbf{X}_1 \hat{\beta}_1)$.

Proof. Let $\tilde{\mathbf{b}} = (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top \mathbf{y}$. Since $\mathbf{I} - \mathbf{M}_2$ is symmetric and idempotent, $\tilde{\mathbf{X}}_1^\top \mathbf{y} = \tilde{\mathbf{X}}_1^\top (\mathbf{I} - \mathbf{M}_2) \mathbf{y} = \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{y}}$, so the two expressions in (a) agree. By Lemma 6.6.1(c),

$$\mathbf{M}\mathbf{y} = \mathbf{M}_2\mathbf{y} + \tilde{\mathbf{X}}_1\tilde{\mathbf{b}} = \mathbf{M}_2\mathbf{y} + \mathbf{X}_1\tilde{\mathbf{b}} - \mathbf{M}_2\mathbf{X}_1\tilde{\mathbf{b}} = \mathbf{X}_1\tilde{\mathbf{b}} + \mathbf{M}_2(\mathbf{y} - \mathbf{X}_1\tilde{\mathbf{b}}).$$

The last term lies in $\mathcal{C}(\mathbf{X}_2)$, so it equals $\mathbf{X}_2\mathbf{c}$ with $\mathbf{c} = (\mathbf{X}_2^\top \mathbf{X}_2)^{-1} \mathbf{X}_2^\top (\mathbf{y} - \mathbf{X}_1\tilde{\mathbf{b}})$. Thus $\mathbf{M}\mathbf{y} = \mathbf{X}_1\tilde{\mathbf{b}} + \mathbf{X}_2\mathbf{c}$. Because $\mathbf{X}$ has full column rank, the coefficient vector representing $\mathbf{M}\mathbf{y}$ is unique, so $\hat{\beta}_1 = \tilde{\mathbf{b}}$ and $\hat{\beta}_2 = \mathbf{c}$. This proves (a) and (c). For (b), $\mathbf{y} - \mathbf{M}\mathbf{y} = (\mathbf{I} - \mathbf{M}_2)\mathbf{y} - \tilde{\mathbf{M}}_1\mathbf{y} = \tilde{\mathbf{y}} - \tilde{\mathbf{X}}_1\tilde{\mathbf{b}}$, which is the residual of the regression of $\tilde{\mathbf{y}}$ on $\tilde{\mathbf{X}}_1$, because $\tilde{\mathbf{X}}_1\tilde{\mathbf{b}} = \tilde{\mathbf{M}}_1\mathbf{y} = \tilde{\mathbf{M}}_1\tilde{\mathbf{y}}$. ■

The proof uses nothing beyond the orthogonal decomposition of the model space. Part (a) contains a small surprise. Residualizing $\mathbf{y}$ is optional, because the part of $\mathbf{y}$ lying in $\mathcal{C}(\mathbf{X}_2)$ is orthogonal to $\tilde{\mathbf{X}}_1$ and drops out of $\tilde{\mathbf{X}}_1^\top \mathbf{y}$ anyway. Residualizing the regressor of interest is essential.

Idea. In a multiple regression, $\hat{\beta}_j$ measures how $\mathbf{y}$ moves with *the part of $\mathbf{x}_j$ that is orthogonal to the other regressors*. A regressor that is almost a linear combination of the others has a tiny orthogonal part, so its coefficient rests on very little information. That is collinearity, seen geometrically.

6.6.2 Consequences

Variances. The FWL formula writes $\hat{\beta}_1$ as a linear function of $\mathbf{y}$. If $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$, then

$$\text{Cov}(\hat{\beta}_1) = \sigma^2 (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1 (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} = \sigma^2 (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1}.$$

Comparing with $\text{Cov}(\hat{\beta}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$ (Theorem 5.5.1) identifies the leading block of the inverse of a partitioned matrix,

$$[(\mathbf{X}^\top \mathbf{X})^{-1}]_{11} = (\mathbf{X}_1^\top (\mathbf{I} - \mathbf{M}_2) \mathbf{X}_1)^{-1} = (\mathbf{X}_1^\top \mathbf{X}_1 - \mathbf{X}_1^\top \mathbf{X}_2 (\mathbf{X}_2^\top \mathbf{X}_2)^{-1} \mathbf{X}_2^\top \mathbf{X}_1)^{-1}, \quad (6.6.1)$$

a Schur-complement identity that Chapter 1 proves by algebra. For a single regressor $\mathbf{x}_j$, with all other columns (including the intercept) in $\mathbf{X}_2$,

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2} = \frac{\sigma^2}{S_{jj}(1 - R_j^2)}, \quad (6.6.2)$$

where $S_{jj} = \sum_i (x_{ij} - \bar{x}_j)^2$ and R_j^2 is the coefficient of determination from regressing $\mathbf{x}_j$ on the other regressors. The second equality is Exercise (B1). The factor $1/(1 - R_j^2)$ is the *variance inflation factor* of Chapter 26.

Centring. Take $\mathbf{X}_2 = \mathbf{1}$. Then $\mathbf{M}_2 = n^{-1}\mathbf{1}\mathbf{1}^\top$ and $\tilde{\mathbf{X}}_1, \tilde{\mathbf{y}}$ are the centred regressors and response. FWL says the slopes in a model with an intercept are the coefficients of a no-intercept regression of centred $\mathbf{y}$ on centred regressors, and part (c) gives the intercept $\bar{y} - \bar{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}_1$. Example (Intercept inside simple regression) was the case $p_1 = 1$.

Detrending and deseasonalizing. The original application was economic time series. Frisch and Waugh (1933) showed that regressing on a time trend together with other variables gives the same slope as first detrending every series. Lovell (1963) proved the analogous statement for seasonal dummy variables, and noted a trap: if only the *response* is seasonally adjusted and the regressors are not, the result is generally wrong. Part (a) of the theorem shows why. What must be residualized is the regressor.

A warning about standard errors. If the residualized regression of $\tilde{\mathbf{y}}$ on $\tilde{\mathbf{X}}_1$ is fitted with ordinary software, the coefficient and residuals are right, but the software divides the residual sum of squares by $n - p_1$ instead of $n - p_1 - p_2$. It does not know that p_2 dimensions were used up in forming $\tilde{\mathbf{y}}$. The reported standard error is too small by the factor $\sqrt{(n - p_1)/(n - p)}$.

Example. (Poverty and murder rates)

For the 50 US states in 2009, regress the murder rate (per 100,000) on the poverty rate, the percentage of single-parent households, and the percentage of the population living in urban areas, with an intercept. The coefficient of poverty is 0.2538. On its own, a simple regression of murder rate on poverty has slope 0.4706, nearly twice as large. Poverty and single parenthood are correlated, and the simple regression gives poverty credit for part of their shared association.

Figure 6.6.1 shows both regressions. Panel (b) is an **added-variable plot**: the residual of murder rate after regression on the other two regressors and the intercept, against the corresponding residual of poverty. By FWL its least squares slope is exactly the multiple-regression coefficient. The listing confirms the equality to machine precision.

The standard error of the coefficient in the full model is 0.0934. Fitting the residualized regression naively gives 0.0905, smaller by the factor 1.0321, which is $\sqrt{(n - 1)/(n - 4)}$ as predicted.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statemachine.load_pandas().data
data = data.drop(index="District of Columbia")      # an extreme point; see the
                                                    # exercises of Section 6.6

y = data["murder"].to_numpy()
x1 = data["poverty"].to_numpy()                    # the coefficient we care
                                                    # about
X2 = np.column_stack([np.ones(len(y)), data["single"], data["urban"]])
X = np.column_stack([x1, X2])

def resid(v, Z):
    """Residual of v after projecting onto C(Z)."""
    coef, *_ = np.linalg.lstsq(Z, v, rcond=None)
    return v - Z @ coef

beta_full, *_ = np.linalg.lstsq(X, y, rcond=None)    # regress y on everything

y_tilde = resid(y, X2)                            # (I - M2) y
```

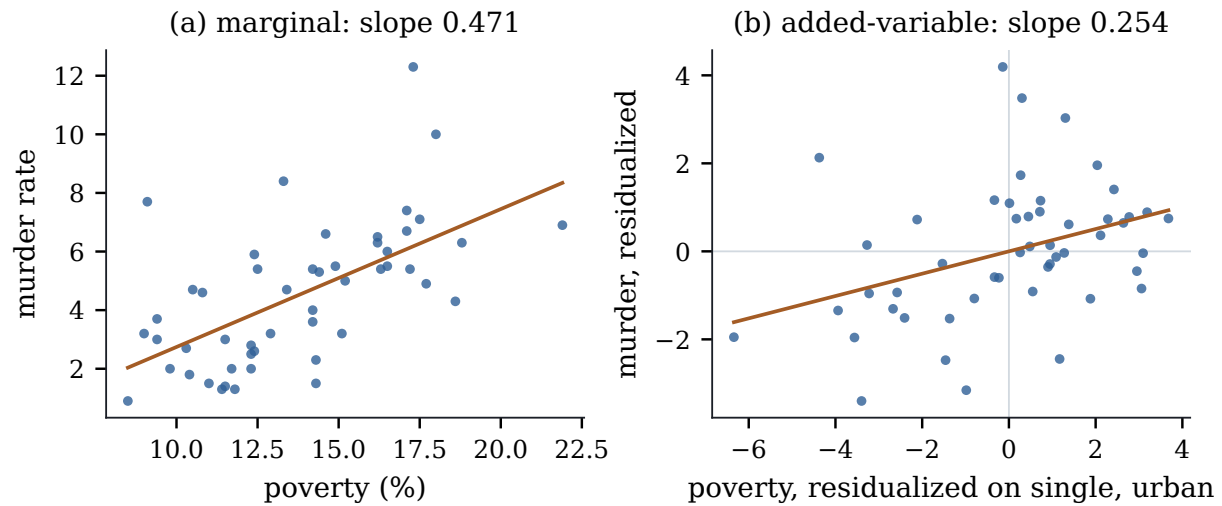


Figure 6.6.1: Murder rate and poverty for 50 US states. (a) The marginal relationship. (b) The added-variable plot: both variables residualized on the single-parent percentage, urban percentage and intercept. The slope in (b) is the multiple-regression coefficient of poverty (Theorem 6.6.2).

```
x_tilde = resid(x1, X2) # (I - M2) x1
beta_fwl = (x_tilde @ y_tilde) / (x_tilde @ x_tilde) # simple regression, no
            intercept

print(f"full regression: {beta_full[0]:.6f}")
print(f"FWL regression: {beta_fwl:.6f}")
```

[Run this code online](#)

Remark (What FWL does not say). FWL is an algebraic identity about fitted coefficients. It says what the number $\hat{\beta}_j$ is: an association with the unexplained part of $\mathbf{x}_j$. It does not say that changing x_j in the world, with the other regressors held fixed, would change the response by $\hat{\beta}_j$. That is a causal claim, and it needs assumptions about how the data arose. Chapter 25 takes up this question, including cases where adding a regressor to $\mathbf{X}_2$ makes the coefficient *less* interpretable rather than more.

6.6.3 Fixed effects are a projection

One of the most widely used estimators in applied statistics is the FWL theorem in disguise. Suppose units $k = 1, \dots, K$ are observed repeatedly: firms over years, patients over visits, schools over cohorts. The analyst wants the effect of time-varying regressors $\mathbf{X}_1$ while allowing each unit its own intercept:

$$E(\mathbf{Y}) = \mathbf{X}_1 \boldsymbol{\beta}_1 + \mathbf{D} \boldsymbol{\alpha},$$

where $\mathbf{D}$ is the $n \times K$ matrix of unit indicators. Fitting this directly means a regression with K extra columns, and K may be in the thousands.

FWL with $\mathbf{X}_2 = \mathbf{D}$ removes that burden. The projection onto $\mathcal{C}(\mathbf{D})$ replaces each entry by its unit's average (Example (The one-way layout)), so $\mathbf{I} - \mathbf{M}_D$ subtracts unit means. Therefore $\hat{\boldsymbol{\beta}}_1$ is the least squares coefficient from regressing the *within-unit deviations* $y_{kt} - \bar{y}_k$ on $\mathbf{x}_{kt} - \bar{\mathbf{x}}_k$, with no indicators at all. This is the **within estimator** of panel-data econometrics. Its standard

error needs the degrees-of-freedom correction described above, with $p_2 = K$, which is large.

Example. (Grunfeld’s investment data)

Grunfeld’s classic panel follows 11 US firms for 20 years, giving $n = 220$ observations of gross investment, market value and capital stock. With a separate intercept for each firm, the coefficients of value and capital are 0.1101 and 0.3100. The same numbers come from demeaning every variable within firm and fitting two columns. Ignoring the firms altogether (one common intercept) gives 0.1145 and 0.2275 instead. That comparison uses variation *between* firms, of the sort “large firms invest more”, which the within estimator deliberately discards.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
df = sm.datasets.grunfeld.load_pandas().data
y = df["invest"].to_numpy()
X1 = df[["value", "capital"]].to_numpy()

# (i) least squares with one indicator column per firm ("dummy variables")
D = pd.get_dummies(df["firm"]).to_numpy(dtype=float)
beta_dummies, *_ = np.linalg.lstsq(np.column_stack([X1, D]), y, rcond=None)

# (ii) FWL: projecting onto C(D) replaces each value by its firm mean,
#         so (I - M_D) subtracts firm means -- the "within" transformation
def demean_by(v, groups):
    v = pd.DataFrame(v)
    return (v - v.groupby(groups.to_numpy()).transform("mean")).to_numpy()

X1_within = demean_by(X1, df["firm"])
y_within = demean_by(y, df["firm"]).ravel()
beta_within, *_ = np.linalg.lstsq(X1_within, y_within, rcond=None)

print("with firm indicators:", beta_dummies[:2])
print("within transformation:", beta_within)
```

[Run this code online](#)

Whether discarding between-unit variation is wise depends on why units differ. That question is taken up for random-effects alternatives in Chapter 32 and Chapter 33.

Added-variable plots are among the most useful diagnostics for multiple regression. They show the evidence for a single coefficient in two dimensions, together with any points that dominate it. They reappear in Chapter 20.

6.6.4 Exercises

6.6.4.1 A. Check your understanding

Exercise A1.

Refit the model of [Example \(Poverty and murder rates\)](#) with the District of Columbia included. How does the poverty coefficient change? Draw the added-variable plot and

locate the District. Relate what you see to its leverage (Example (High-leverage states)).

6.6.4.2 B. Practice

Exercise B1.

Let $\mathbf{x}_j$ be a column of $\mathbf{X}$, let $\mathbf{X}_{(j)}$ be $\mathbf{X}$ with that column removed, and assume $\mathbf{1} \in \mathcal{C}(\mathbf{X}_{(j)})$. Show that $\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2 = S_{jj}(1 - R_j^2)$, where R_j^2 is the R^2 from regressing $\mathbf{x}_j$ on $\mathbf{X}_{(j)}$. Deduce [Eq. 6.6.2](#).

Solution. Since $\mathbf{1} \in \mathcal{C}(\mathbf{X}_{(j)})$, the vector $(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j$ is the residual from regressing $\mathbf{x}_j$ on $\mathbf{X}_{(j)}$. Its squared length is that regression's residual sum of squares, which equals $\text{SST}(1 - R_j^2)$ with $\text{SST} = S_{jj}$ by the definition of R^2 . Substituting into the variance formula from FWL gives [Eq. 6.6.2](#).

Exercise B2.

In [Theorem 6.6.2](#), regressing $\mathbf{y}$ (not $\tilde{\mathbf{y}}$) on $\tilde{\mathbf{X}}_1$ gives the coefficient $\hat{\beta}_1$. Show that its residual vector is $(\mathbf{I} - \mathbf{M})\mathbf{y} + \mathbf{M}_2\mathbf{y}$, so the residuals are generally wrong even though the coefficient is right.

Solution. The fitted vector of the regression of $\mathbf{y}$ on $\tilde{\mathbf{X}}_1$ is $\tilde{\mathbf{M}}_1\mathbf{y}$. By [Lemma 6.6.1](#), $\mathbf{y} - \tilde{\mathbf{M}}_1\mathbf{y} = \mathbf{y} - (\mathbf{M} - \mathbf{M}_2)\mathbf{y} = (\mathbf{I} - \mathbf{M})\mathbf{y} + \mathbf{M}_2\mathbf{y}$. The extra term $\mathbf{M}_2\mathbf{y}$ is zero only if $\mathbf{y} \perp \mathcal{C}(\mathbf{X}_2)$.

Exercise B3.

For a single regressor $\mathbf{x}_j$, the *partial correlation* of $\mathbf{y}$ and $\mathbf{x}_j$ given the other regressors is the sample correlation of $\tilde{\mathbf{y}}$ and $\tilde{\mathbf{x}}_j$ (both residualized on $\mathbf{X}_{(j)}$, which contains $\mathbf{1}$). Let $t = \hat{\beta}_j/\text{se}(\hat{\beta}_j)$ with $\text{se}(\hat{\beta}_j)^2 = \text{SSE}/[(n - p)\|\tilde{\mathbf{x}}_j\|^2]$. Show that the squared partial correlation equals $t^2/(t^2 + n - p)$.

Solution. Write $\tilde{\mathbf{x}} = \tilde{\mathbf{x}}_j$ and $\tilde{\mathbf{y}} = \tilde{\mathbf{y}}$. Both have mean zero because $\mathbf{1} \in \mathcal{C}(\mathbf{X}_{(j)})$, so their correlation is $r = \tilde{\mathbf{x}}^\top \tilde{\mathbf{y}}/(\|\tilde{\mathbf{x}}\|\|\tilde{\mathbf{y}}\|)$. By FWL, $\hat{\beta}_j = \tilde{\mathbf{x}}^\top \tilde{\mathbf{y}}/\|\tilde{\mathbf{x}}\|^2$, and the full-model residual sum of squares equals that of the residualized regression: $\text{SSE} = \|\tilde{\mathbf{y}}\|^2 - (\tilde{\mathbf{x}}^\top \tilde{\mathbf{y}})^2/\|\tilde{\mathbf{x}}\|^2 = \|\tilde{\mathbf{y}}\|^2(1 - r^2)$. Hence

$$t^2 = \frac{\hat{\beta}_j^2 (n - p) \|\tilde{\mathbf{x}}\|^2}{\text{SSE}} = \frac{(n - p) r^2}{1 - r^2},$$

and solving gives $r^2 = t^2/(t^2 + n - p)$.

6.6.4.3 C. Going deeper

Exercise C1.

Drop the full-rank assumption in [Theorem 6.6.2](#). Show that $\mathbf{b}_1$ solves the normal equations of the regression of $\tilde{\mathbf{y}}$ on $\tilde{\mathbf{X}}_1$ iff there exists $\mathbf{b}_2$ such that $(\mathbf{b}_1^\top, \mathbf{b}_2^\top)^\top$ is a least squares estimate in the full model.